

INI CET & AIIMS PG PEDIATRICS

TOP 20 HIGH-YIELD TOPICS — AUDIT EDITION

2015 – 2025 · Verified Recalls · One Topic Per Page

Topics ranked by confirmed presence in recall papers from all 20 sessions (2015-2025)

Changes from v1.0: Kawasaki Disease replaced by Cystic Fibrosis; Childhood Asthma replaced by Neonatal RDS

#	Topic	#	Topic
01	Neonatal Jaundice & Biliary Atresia	11	Growth, Development & Milestones
02	Febrile Seizures	12	Cystic Fibrosis
03	Nephrotic Syndrome	13	Dengue Fever in Children
04	Lysosomal Storage Disorders	14	Thalassemia & Hemoglobin Disorders
05	Tetralogy of Fallot	15	Congenital Hypothyroidism
06	Severe Acute Malnutrition (SAM)	16	Acute Rheumatic Fever & RHD
07	Vitamin D Deficiency Rickets	17	Henoch-Schonlein Purpura
08	Primary Immunodeficiency Disorders	18	Neurocutaneous Syndromes
09	Meningitis in Children	19	Vaccination Schedule & NIP
10	Childhood Tuberculosis	20	Neonatal RDS & Neonatal Infections

Topic 01

Neonatal Jaundice & Biliary Atresia

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024,Nov 2024,May 2025 (HIDA scan question confirmed)

Exam Frequency (Audited): Confirmed in EVERY session (2015-2025); one of the TOP 3 most asked pediatrics topics

OVERVIEW

Neonatal jaundice affects 60% of term and 80% of preterm neonates. Biliary atresia (BA) is the leading cause of obstructive neonatal cholestasis and the most common indication for liver transplantation in children. Both conditions are perennially tested — jaundice for basic science and BA for clinical management decisions.

KEY FACTS FOR EXAM

- Physiological jaundice: appears Day 2-3, peaks Day 4-5, resolves by Day 14 (term); due to high RBC turnover + immature liver conjugation
- Pathological jaundice: onset <24 hours; rate >0.5 mg/dL/hr; direct bilirubin >2 mg/dL; or persists >14 days (term)
- Most common cause of neonatal jaundice overall: physiological; most common pathological cause: ABO incompatibility
- Biliary atresia: most common type III (>90%); progressive obliteration of extrahepatic bile ducts; Kasai portoenterostomy is primary surgery
- HIDA scan (hepatobiliary scintigraphy): no excretion into gut = biliary atresia (not diagnostic alone); confirms with liver biopsy
- Kasai procedure: best outcomes if done before 60 days of age; otherwise liver transplant is primary option
- Jaundice Day 1 = pathological until proven otherwise; commonest cause = Rh/ABO incompatibility
- Kernicterus: basal ganglia + hippocampus + subthalamic nucleus; MRI = T1 hyperintensity in globus pallidus

CLINICAL FEATURES

Kramer's zones	Zone 1=face only (5 mg/dL) → Zone 5=palms/soles (>20 mg/dL)
Biliary atresia	Progressive jaundice from 2-3 weeks; pale acholic stools; dark urine; hepatomegaly
ABO incompatibility	Jaundice Day 1-2; positive DAT; spherocytes on PBS; usually milder than Rh
Rh incompatibility	Severe hemolysis; hydrops fetalis; positive DAT; no spherocytes
G6PD deficiency	Jaundice triggered by naphthalene/infections/drugs; Heinz bodies; Coombs negative
Phototherapy	Blue light 460-490 nm; converts bilirubin to lumirubin (water-soluble); distance 15-20 cm

KEY INVESTIGATIONS

- Total/direct bilirubin: plot on Bhutani nomogram by age in hours
- DAT (Direct Coombs): positive in immune hemolysis (Rh, ABO); negative in G6PD, physiological
- HIDA scan: gold standard to distinguish BA from neonatal hepatitis (excretion rules out BA)
- Liver biopsy: confirms BA; shows bile duct proliferation, portal fibrosis, bile plugs

EXAM TIP

Confirmed 2025 question: infant with 3 months progressive jaundice, pale stools, HIDA scan no excretion → answer was Kasai portoenterostomy (NOT liver transplant as first step). Key: (1) Pale/acholic stools = always investigate for BA. (2) HIDA no excretion = suspect BA; confirm with biopsy. (3) Kasai before 60 days = best outcome. (4) Phototherapy wavelength 460-490 nm (NOT UV). (5) ABO → spherocytes; Rh → no spherocytes.

Topic 02

Febrile Seizures

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Asked in 80%+ of all sessions; classification and management are the most tested sub-topics

OVERVIEW

Febrile seizures are the most common seizure disorder in childhood, occurring in 2-5% of children aged 6 months to 6 years. Questions focus on the simple vs complex distinction, LP criteria, prophylaxis decisions, and acute management. Consistently tested across all AIIMS PG and INI CET sessions.

KEY FACTS FOR EXAM

- Age group: 6 months - 6 years; peak 12-18 months; M > F slightly
- Simple FS: generalized, <15 min, no recurrence within 24 h, neurologically normal — ALL three must be met
- Complex FS: focal OR duration >15 min OR >1 episode in 24 h — ANY ONE criterion makes it complex
- Febrile Status Epilepticus: seizure >30 min; treat with IV/rectal benzodiazepine urgently
- Risk of recurrence: ~33%; lower age + family history + low-grade fever at onset = higher risk
- Risk of subsequent epilepsy: 2-3% (simple FS); 4-6% (complex); general population baseline 1%
- LP: needed if <12 months (meningism signs unreliable); partially vaccinated; prior antibiotics given
- No routine EEG after simple FS — EEG does not predict recurrence or epilepsy risk

CLINICAL FEATURES

Onset	Rapidly rising temperature; seizure usually occurs at rising phase, not peak
Simple FS	Generalized tonic-clonic; <5 minutes usually; brief post-ictal period; returns to baseline
Complex FS	Todd's paresis may follow focal seizure; longer duration; repeat episodes concern for meningitis
Post-ictal	Brief drowsiness normal in simple FS; prolonged altered consciousness → investigate
Temperature	Usually >38.5°C; viral URTIs most common trigger (HHV-6 = roseola most common viral cause)
Family history	Autosomal dominant tendency; if parent had FS, child has 2-3x higher risk

KEY INVESTIGATIONS

- No routine blood tests or imaging for simple FS in a vaccinated, well-appearing child
- LP criteria: age <12 months, partially vaccinated, prior antibiotics, meningeal signs
- EEG: NOT indicated for simple FS; consider for complex FS or subsequent afebrile seizures
- Blood glucose: always check to exclude hypoglycemic seizure

EXAM TIP

Critical distinctions: (1) Complex FS = ANY ONE of: focal, >15 min, >1 in 24h. (2) Drug for prolonged FS: diazepam (IV/rectal) or midazolam (buccal/intranasal). (3) Prophylaxis: NOT recommended for simple FS; intermittent diazepam used for complex/recurrent in selected cases. (4) LP threshold lower in infants <12 months. (5) HHV-6 (roseola) = most common viral cause of FS. (6) Valproate: used for prophylaxis if needed, NOT phenobarbitone.

Nephrotic Syndrome in Children

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,Nov 2024,May 2025 (diagnostic criteria question confirmed)

Exam Frequency (Audited): Confirmed every session; SBP organism, steroid protocol, and biopsy indications are most frequently tested

OVERVIEW

Idiopathic nephrotic syndrome is the most common glomerular disease in children. Minimal change disease (MCD) accounts for 85-90% in children aged 1-8 years. The INI CET 2025 session confirmed a question on definitive diagnostic criteria. Questions span diagnosis, steroid protocol, complications, and biopsy indications.

KEY FACTS FOR EXAM

- Diagnostic criteria: Proteinuria ≥ 40 mg/m²/hr OR spot urine protein:creatinine > 2 OR 3+ dipstick $\times 3$ days + edema + hypoalbuminemia < 2.5 g/dL
- Edema: periorbital (morning), dependent (evening); ascites, pleural effusion, scrotal/labial edema in severe disease
- MCD: most common; steroid-sensitive; foot process effacement on EM; normal light microscopy; good prognosis
- FSGS: steroid-resistant; poor prognosis; biopsy essential; secondary causes include HIV, obesity, sickle cell
- First episode treatment: prednisolone 2 mg/kg/day (max 60 mg) $\times 4$ weeks, then 1.5 mg/kg alternate days $\times 4$ weeks
- Steroid-sensitive NS (SSNS): complete remission within 4 weeks; steroid-resistant NS (SRNS): no remission after 4-6 weeks
- Most common infection in NS: spontaneous bacterial peritonitis (SBP) — *Streptococcus pneumoniae*
- Renal vein thrombosis: most specific thrombotic complication; due to urinary loss of antithrombin III

CLINICAL FEATURES

Edema	Pitting; begins periorbital (morning) → ascites → pleural effusion → anasarca
Urine	Frothy urine; reduced output during relapse; Maltese cross on polarized microscopy (lipiduria)
Blood pressure	Normal in MCD; hypertension in MPGN, FSGS suggests different pathology
SBP	Abdominal pain + fever + peritoneal signs; <i>Strep. pneumoniae</i> most common organism
Volume depletion	Cold extremities, tachycardia, reduced urine output — hypovolemia despite edema
Thrombosis	Renal vein thrombosis; pulmonary embolism; cerebral sinus thrombosis

KEY INVESTIGATIONS

EXAM TIP

- Spot urine P:Cr ratio: > 2 = nephrotic range (first morning sample most accurate)
 - Serum albumin < 2.5 g/dL; cholesterol elevated; C3/C4 normal in MCD (low in MPGN, lupus)
 - Renal biopsy: age < 1 yr or > 12 yr; SRNS; frequent relapse; steroid-dependent; atypical features
 - EM: foot process effacement (MCD); subepithelial humps (PSGN); subendothelial deposits (MPGN)
- Confirmed May 2025 question tested definitive diagnostic criteria for NS. Key facts: (1) SBP organism = *Streptococcus pneumoniae* (encapsulated, complement consumed in NS). (2) Low C3 in NS = not MCD — think MPGN or lupus. (3) MCD on EM = foot process effacement (podocytes). (4) FSGS on LM = segmental sclerosis + hyalinosis. (5) Pneumococcal vaccine mandatory in NS children. (6) Cyclophosphamide used for frequently relapsing NS.*

Lysosomal Storage Disorders

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024 (Gaucher question confirmed multiple sessions)

Exam Frequency (Audited): Confirmed in virtually every session; Gaucher, Tay-Sachs, Niemann-Pick, Hurler most tested

OVERVIEW

Lysosomal storage disorders (LSDs) are a classic AIIMS/INI CET topic due to their high-density factual content. The exam consistently tests enzyme-substrate-clinical feature correlations and distinguishing between conditions that look similar (e.g., Tay-Sachs vs Niemann-Pick; Gaucher vs other HSM causes).

KEY FACTS FOR EXAM

- Gaucher disease: beta-glucocerebrosidase deficiency; Gaucher cells = 'crumpled tissue-paper' cytoplasm; HSM; NO cherry-red spot
- Gaucher Type 1 (non-neuropathic): most common; 'Erlenmeyer flask' femur; bone pain; avascular necrosis; treatable with ERT
- Tay-Sachs: Hex-A deficiency; cherry-red spot; NO hepatosplenomegaly; progressive neuro; macrocephaly; Ashkenazi Jews
- Niemann-Pick Type A: sphingomyelinase deficiency; cherry-red spot + HSM; 'foam cells'; fatal by 3-5 years
- Hurler (MPS-IH): alpha-L-iduronidase; gargoye facies; cloudy cornea; cardiac valve disease; dysostosis multiplex; HSCT curative
- Fabry disease: alpha-galactosidase A; X-linked; angiokeratoma (bathing-trunk); painful crises; renal failure; treated with ERT
- Pompe disease (GSD Type II): acid maltase (alpha-glucosidase); hypertrophic CMP; hypotonia; 'snowstorm' CXR in infants
- NBT/DHR test: CGD (not LSD); Enzyme assay on leukocytes/fibroblasts = gold standard for LSDs

CLINICAL FEATURES

Cherry-red spot	Tay-Sachs, Niemann-Pick A, Sandhoff, GM1 gangliosidosis, Farber disease — NOT in Gaucher
Hepatosplenomegaly	Gaucher, Niemann-Pick, Hurler — NOT in Tay-Sachs (no visceral involvement)
Cloudy cornea	Hurler (MPS-IH), Scheie, Maroteaux-Lamy — NOT in Gaucher or Tay-Sachs
Angiokeratoma	Fabry disease; small dark-red papules; bathing-trunk distribution; navel, scrotum, thighs
Dysostosis multiplex	MPS (Hurler, Hunter), GM1 — NOT in Gaucher or Tay-Sachs
Psychiatric symptoms	Late-onset Tay-Sachs; Niemann-Pick Type C (vertical gaze palsy + ataxia + dementia)

KEY INVESTIGATIONS

EXAM TIP

- Enzyme assay (leukocytes or fibroblasts): gold standard for all LSDs
 - Chitotriosidase: markedly elevated in Gaucher (monitoring marker); not disease-specific
 - Urine GAGs: elevated in MPS disorders (Hurler, Hunter); spot test + quantification
 - BM biopsy: Gaucher cells (crumpled tissue paper); foam cells (Niemann-Pick) — supplementary
- Master this comparison table: Gaucher = no cherry-red spot + HSM + bone disease. Tay-Sachs = cherry-red spot, NO HSM, macrocephaly, Ashkenazi Jews. Niemann-Pick A = cherry-red spot + HSM + foam cells. Hurler = corneal clouding + gargoye facies + dysostosis multiplex. Treatment: Gaucher Type 1 = imiglucerase (ERT); Fabry = agalsidase; Hurler = HSCT (curative if done early). Hunter (MPS-II) = X-linked (no corneal clouding unlike Hurler).*

Tetralogy of Fallot (TOF)

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Most tested cyanotic CHD across all sessions; Tet spells and X-ray/ECG findings most commonly asked

OVERVIEW

Tetralogy of Fallot is the most common cyanotic congenital heart disease beyond infancy (10% of all CHD). It represents the prototype for understanding right-to-left shunts, cyanotic spells, and corrective surgery. Every AIIMS PG and INI CET paper has had at least one question on TOF.

KEY FACTS FOR EXAM

- Four components: (1) Large non-restrictive VSD, (2) RVOTO (infundibular ± valvular), (3) Overriding aorta, (4) RVH
- Embryological basis: anterosuperior displacement of infundibular (conal) septum
- Degree of cyanosis depends on severity of RVOTO — more obstruction = more R-to-L shunt = more cyanosis
- Tet spell (hypercyanotic spell): peak age 2-4 months; agitation → paroxysmal hyperpnea → cyanosis → LOC/death
- Tet spell management: knee-chest position → O₂ → IV morphine → IV propranolol → IV phenylephrine
- X-ray: boot-shaped heart (coeur en sabot); absent MPA segment; right aortic arch in 25%
- ECG: right axis deviation (RAD) + right ventricular hypertrophy (RVH)
- Louder murmur = less severe obstruction (more blood passing through RVOTO = louder ejection systolic murmur)

CLINICAL FEATURES

Cyanosis	Central; appears at 1-3 months as PDA closes; clubbing develops after 6 months
Murmur	Ejection systolic at ULSE; from RVOTO not VSD; P2 soft/absent; single S2
Squatting	Increases SVR → reduces R-to-L shunt → improves O ₂ saturation; seen in older children
Tet spell	Crying/feeding triggers; cyanosis worsens; loss of murmur indicates severe obstruction
Pink TOF	Minimal RVOTO; predominantly L-to-R shunt; not cyanotic at rest; murmur louder
Polycythemia	Compensatory; hyperviscosity → thrombosis risk; keep Hb <20 g/dL

KEY INVESTIGATIONS

- Echo: diagnostic; shows all 4 components; essential to define coronary artery anatomy pre-operatively
- CXR: boot-shaped heart; oligemic lung fields (reduced pulmonary blood flow)
- ECG: RVH + RAD + right atrial enlargement
- ABG during spell: severe hypoxemia + metabolic acidosis

EXAM TIP

Must-know: (1) Loudest murmur = LEAST severe disease. (2) First step in Tet spell = knee-chest (squatting equivalent). (3) Propranolol reduces infundibular spasm. (4) Phenylephrine increases SVR (pushes blood to lungs). (5) Pentalogy of Fallot = TOF + ASD. (6) MAPCA (major aortopulmonary collateral arteries) in TOF with absent pulmonary valve = complex repair. (7) Definitive repair: VSD closure + RVOTO relief; BT shunt (subclavian-PA) is palliative.

Topic 06

Severe Acute Malnutrition (SAM)

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024,2025

Exam Frequency (Audited): Confirmed every session; WHO 10-step protocol, F-75/F-100/RUTF, and refeeding syndrome most tested

OVERVIEW

SAM carries high mortality (20-30% without treatment). The WHO 10-step protocol, F-75/F-100 rationale, RUTF, and MUAC thresholds are classic exam content. IAP and WHO guidelines are updated periodically — the exam tests both clinical criteria and management nuances.

KEY FACTS FOR EXAM

- SAM criteria: W/H Z-score <-3 SD OR MUAC <11.5 cm OR bilateral pitting edema (any ONE criterion sufficient)
- MAM: MUAC 11.5-12.5 cm; W/H Z-score -2 to -3 SD; managed with supplementary feeding
- Marasmus: severe wasting; 'old man face'; 'baggy pants' sign; alert but hungry; no edema
- Kwashiorkor: bilateral edema; flaky-paint dermatosis; hair changes (flag sign = alternating bands); fatty liver; irritable
- WHO 10 steps: hypoglycemia → hypothermia → dehydration → electrolytes → infections → micronutrients → feeding → growth → emotional/sensory → follow-up
- F-75: used in stabilization (low protein, low energy); prevents refeeding syndrome; NOT F-100 in stabilization
- F-100 / RUTF: rehabilitation phase; RUTF = Ready-to-Use Therapeutic Food (peanut paste based)
- Refeeding syndrome: hypophosphatemia + hypokalemia + hypomagnesemia; occurs first 48-72 hours of refeeding

CLINICAL FEATURES

MUAC	Mid-upper arm circumference; does NOT need age; <11.5 cm = SAM; best field screening tool
Bilateral edema	Press dorsum of foot ≥ 3 seconds; pitting = kwashiorkor component; grade 1-3
Appetite test	RUTF given to child; if eats = outpatient (CMAM); refuses = inpatient
Skin	Flaky-paint dermatosis (kwashiorkor); skin peeling, erosions, dark patches
Hair	Flag sign = bands of pale/dark hair; easily pluckable; reddish discoloration in kwashiorkor
Metabolic complications	Hypoglycemia (<3 mmol/L); hypothermia ($<35^{\circ}\text{C}$ axillary); electrolyte imbalances

KEY INVESTIGATIONS

- Blood glucose: check immediately (hypoglycemia = silent in SAM; may present as hypothermia only)
- Electrolytes: Na (hyponatremia common), K (hypokalemia), Mg (hypomagnesemia), PO₄ (refeeding risk)
- Avoid iron in stabilization phase (iron worsens oxidative stress; give only in rehabilitation)
- HIV/TB screen in endemic areas; stool R/E; blood culture if sepsis suspected

EXAM TIP

Classic exam traps: (1) Use F-75 NOT F-100 in stabilization — F-100 causes refeeding syndrome. (2) Antibiotics given to ALL SAM children even without obvious infection (amoxicillin +/- metronidazole). (3) Iron only in rehabilitation phase (week 3 onwards). (4) MUAC: preferred field tool — no need to know exact age. (5) Therapeutic feeding targets: 130-200 kcal/kg/day in rehabilitation. (6) Breastfed infants: continue breastfeeding.

Vitamin D Deficiency Rickets

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Confirmed in most sessions; biochemical profile, X-ray findings, and resistant rickets are most tested

OVERVIEW

Nutritional rickets due to vitamin D deficiency remains the most common metabolic bone disease in Indian children. The cascade from low vitamin D → low calcium → PTH surge → phosphaturia → impaired mineralization is consistently tested. The exam distinguishes nutritional rickets from hereditary forms.

KEY FACTS FOR EXAM

- Pathophysiology: low 25-OH vitamin D → low serum calcium → secondary hyperparathyroidism → phosphaturia → hypophosphatemia → impaired mineralization
- Classic biochemistry: Ca low/normal-low, PO₄ low, ALP very high (most sensitive marker), PTH high, 25-OH vit D low
- Radiological hallmark: cupping, fraying, widening of metaphysis — best seen at wrist (distal radius/ulna)
- Craniotabes: earliest clinical sign (ping-pong ball sensation on skull); occurs in first few months
- Rachitic rosary: beading at costochondral junctions; soft, tender; felt on rib cage
- Treatment: Vitamin D 60,000 IU/week × 8-12 weeks (oral); then maintenance 400-800 IU/day
- X-linked hypophosphatemic rickets (XLH): most common hereditary form; normal Ca, low PO₄, normal PTH, high FGF23; needs phosphate + calcitriol
- Vitamin D-dependent rickets Type I (VDDR-I): CYP27B1 mutation; low 1,25(OH)₂D; responds to calcitriol

CLINICAL FEATURES

Head	Craniotabes (earliest); frontal bossing; delayed anterior fontanelle closure; delayed dentition
Chest	Rachitic rosary; Harrison's sulcus (horizontal groove at diaphragm insertion); pigeon chest
Limbs	Genu varum (bow legs, <2 years); genu valgum (knock knees, 2-5 years); wrist widening
Muscle	Hypotonia; delayed motor milestones; ligament laxity; waddling gait
Dental	Enamel hypoplasia; multiple caries; delayed eruption
Tetany	Carpopedal spasm; Chvostek sign; Trousseau sign; laryngospasm — occurs with acute hypocalcemia

KEY INVESTIGATIONS

- ALP: most consistently elevated; use as screening and monitoring marker
- 25-OH vitamin D: <20 ng/mL = deficiency; <12 ng/mL = severe deficiency
- X-ray wrist: cupping + fraying + widening of growth plate (zone of provisional calcification widened)
- PTH: elevated (secondary hyperparathyroidism); distinguishes from primary phosphate wasting disorders

EXAM TIP

Key discriminators: (1) ALP = MOST sensitive lab marker for rickets (always elevated). (2) First clinical sign = craniotabes. (3) Best X-ray site = wrist (distal radius/ulna metaphysis). (4) XLH (X-linked hypophosphatemic): phosphate wasting, FGF23 excess, normal PTH, normal 25-OH D — does NOT respond to vitamin D alone. (5) VDDR-I: responds to calcitriol (not cholecalciferol). (6) VDDR-II: end-organ resistance; alopecia; very high calcitriol.

Primary Immunodeficiency Disorders

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024 (CGD, XLA, SCID confirmed multiple sessions)

Exam Frequency (Audited): Confirmed 80%+ of sessions; NBT/DHR test, delayed cord separation, XLA features most tested

OVERVIEW

Primary immunodeficiency disorders (PIDs) are tested through clinical pattern recognition: the type of organism, age of onset, and associated features point to the specific defect. AIIMS/INI CET consistently tests CGD, XLA, SCID, LAD, Chediak-Higashi, and DiGeorge syndrome.

KEY FACTS FOR EXAM

- XLA (Bruton's): X-linked; BTK mutation; absent B cells (CD19-); very low/absent IgG/A/M; onset after 6 months (maternal Ab wane)
- SCID: absent T + B cells; from birth; failure to thrive + severe infections; graft-vs-host from maternal T cells or transfusions
- CGD: NADPH oxidase defect; X-linked (gp91) or AR (p22,p47,p67); catalase-positive organisms (Staph, Aspergillus, Serratia, Nocardia)
- LAD-1: CD18 (beta-2 integrin) deficiency; delayed umbilical cord separation (>21 days); no pus formation; leukocytosis without infection
- Chediak-Higashi: AR; LYST gene; giant granules on PBS; oculocutaneous albinism; recurrent pyogenic infections; NK cell dysfunction
- DiGeorge: 22q11.2 deletion; T cell deficiency; absent thymic shadow on CXR; hypocalcemia; conotruncal heart defects (TOF, IAA, truncus)
- Wiskott-Aldrich: X-linked; WAS gene; triad = eczema + thrombocytopenia (small platelets) + immunodeficiency
- NBT test (nitroblue tetrazolium): turns blue in normal; STAYS YELLOW in CGD — gold standard screening; confirm with DHR flow cytometry

CLINICAL FEATURES

B cell defects	Recurrent encapsulated bacterial infections; onset after 6 months; no lymph nodes/tonsils (XLA)
T cell defects	Viral, fungal, PCP, mycobacterial from birth; failure to thrive; chronic diarrhea
SCID	Most severe; maternal T cell engraftment causes GVHD; do NOT give live vaccines or irradiated blood
CGD	Catalase+ organism infections; liver abscesses; lymphadenitis; granuloma formation; NBT negative
LAD-1	Leukocytosis without pus; wound non-healing; delayed cord separation hallmark; gingivitis
Complement C5-C9 deficiency	Recurrent Neisseria (meningococcal/gonococcal); poor opsonization

KEY INVESTIGATIONS

- XLA: flow cytometry (absent CD19+ B cells); serum IgG/A/M very low (<2 mg/dL each)
- CGD: NBT test (no blue color); DHR test (no oxidative burst); confirmed by genetic mutation analysis
- SCID: lymphopenia; absent T cell mitogens response; absent adenosine deaminase (ADA deficiency type)
- LAD: flow cytometry (absent CD18 and CD11b on neutrophils)

EXAM TIP

Pattern approach: (1) No Igs + no B cells after 6 months = XLA. (2) No T cells from birth = SCID. (3) NBT stays yellow = CGD. (4) Cord falls off >21 days = LAD. (5) Giant granules + albinism = Chediak-Higashi. (6) 22q11 = DiGeorge (CATCH-22: Cardiac, Abnormal facies, Thymic aplasia, Cleft palate, Hypocalcemia). (7) SCID treatment: HSCT (curative); gene therapy for ADA-SCID. (8) CGD prophylaxis: TMP-SMX + itraconazole.

Meningitis in Children

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024 (CSF interpretation, organism by age confirmed)

Exam Frequency (Audited): Confirmed in most sessions; CSF analysis interpretation and organism-by-age are the most tested aspects

OVERVIEW

Bacterial meningitis is a medical emergency. Questions test CSF interpretation, organism identification by age group, antibiotic choice, adjunctive dexamethasone indications, and complication recognition. TB meningitis staging and management are also regularly tested.

KEY FACTS FOR EXAM

- Neonates (0-3 months): GBS + E. coli + Listeria → ampicillin + gentamicin (or ampicillin + cefotaxime)
- Infants and children: N. meningitidis + S. pneumoniae + H. influenzae → ceftriaxone (± vancomycin for resistant Strep)
- CSF bacterial meningitis: WBC >1000 (PMNs), glucose <40 mg/dL, protein >150 mg/dL, high opening pressure
- CSF viral meningitis: WBC 10-500 (lymphocytes), glucose normal, protein slightly raised
- CSF TB meningitis: lymphocytosis, very low glucose (<45), very high protein, pellicle, xanthochromic
- Dexamethasone: reduces sequelae in H. influenzae and S. pneumoniae; must be given WITH or BEFORE first antibiotic dose
- Most common sequela: sensorineural hearing loss — all children should have audiological follow-up after meningitis
- Waterhouse-Friderichsen: N. meningitidis septicemia → bilateral adrenal hemorrhage + DIC + fulminant purpura

CLINICAL FEATURES

Classical triad	Fever + headache + neck stiffness; photophobia; phonophobia; kernig's sign; brudzinski's sign
Infants	Bulging fontanelle; high-pitched cry; poor feeding; seizures — meningismus signs ABSENT in neonates
Purpuric rash	Non-blanching petechiae → purpura = meningococcemia; give benzylpenicillin IMMEDIATELY
Kernig's sign	Flex hip 90° → cannot extend knee >135° due to pain; positive in meningitis
Brudzinski's sign	Passive neck flexion → involuntary hip and knee flexion
SIADH	Complication of meningitis; hyponatremia; fluid restriction required

KEY INVESTIGATIONS

- CT before LP: only if papilloedema, GCS <13, focal deficit, prolonged seizures, immunocompromised
- CSF: Gram stain + culture + WBC + glucose (with simultaneous blood glucose) + protein
- Blood culture: obtain BEFORE antibiotics if LP is delayed; do not delay antibiotics to do CT
- Serum procalcitonin: >0.5 ng/mL supports bacterial over viral meningitis

EXAM TIP

Exam essentials: (1) Most common sequela = sensorineural hearing loss. (2) Dexamethasone = give WITH FIRST dose of antibiotics (not after). (3) Empiric Abx: neonates = ampicillin + gentamicin; children = ceftriaxone ± vancomycin. (4) TB meningitis stages: Stage I (alert, no deficit) → II (confusion, CN palsy) → III (coma). (5) TB meningitis = 2HRZE + 10HR = 12 months total; add corticosteroids. (6) Pneumococcal meningitis: add vancomycin empirically (PRP strains).

Topic 10

Childhood Tuberculosis

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024,2025

Exam Frequency (Audited): Confirmed every session; drug side effects, miliary TB, TST interpretation most tested

OVERVIEW

Pediatric TB is predominantly lymph-node based or presents as primary complex. Children are paucibacillary making diagnosis challenging. Drug side effects, TST interpretation, BCG vaccine facts, and miliary TB are classic high-yield areas that recur in nearly every AIIMS PG and INI CET paper.

KEY FACTS FOR EXAM

- Ghon focus (primary lung lesion) + regional lymphadenopathy = Ghon complex; both together = primary complex
- TST (Mantoux): read at 48-72 hours; measure transverse induration (not erythema); >10 mm positive in BCG-vaccinated
- IGRA (QuantiFERON): not affected by BCG vaccination; preferred in BCG-vaccinated populations for latent TB
- Miliary TB: hematogenous dissemination; 'snowstorm' CXR; choroidal tubercles on fundoscopy; most common in <2 years
- Treatment DS-TB: 2HRZE + 4HR (6 months); TB meningitis/miliary = 2HRZE + 10HR (12 months total); steroids mandatory in TBM
- Drug side effects: Isoniazid = hepatotoxicity + peripheral neuropathy; Rifampicin = orange urine + enzyme inducer
- Ethambutol: retrobulbar neuritis (colour vision loss: red-green); avoid <6 years (cannot report visual symptoms)
- Pyrazinamide: hyperuricemia + arthralgia + hepatotoxicity; Streptomycin: ototoxicity + nephrotoxicity

CLINICAL FEATURES

Primary TB	Usually asymptomatic; hilar lymphadenopathy; can cause wheeze/stridor from airway compression
Miliary TB	Fever + weight loss; CXR 'snowstorm'; choroidal tubercles (pathognomonic on fundoscopy)
TB meningitis	Subacute course; basal meningitis → CN palsies (CN VI most common); hydrocephalus; vasculitis
TB lymphadenitis	Cervical nodes most common; matting; collar-stud abscess; scrofuloderma
Pott's disease	Thoracic spine most common; angular kyphosis (gibbus); paraplegia from cord compression
BCG	Protects against miliary TB and TBM (disseminated disease) but not primary pulmonary infection

KEY INVESTIGATIONS

EXAM TIP

- TST: read transverse induration at 48-72h; ≥10 mm positive (immunocompetent, BCG vaccinated)
 - GeneXpert MTB/RIF: rapid diagnosis; detects rifampicin resistance simultaneously
 - Gastric aspirate / induced sputum: for bacteriological confirmation (children cannot expectorate)
 - CXR: hilar adenopathy, primary complex, miliary pattern, pleural effusion, cavitation
- Drug side effects (most tested): (1) Isoniazid = hepatotoxicity (most common), peripheral neuropathy (give B6). (2) Rifampicin = orange urine, enzyme inducer (reduces OCP efficacy, warfarin). (3) Ethambutol = retrobulbar neuritis (red-green colour blindness) — avoid <6 years. (4) Pyrazinamide = hyperuricemia, arthralgia. (5) Best biopsy site in miliary TB = liver or bone marrow (not lung). (6) BCG strain: Danish 1331 or Tokyo 172.*

Topic 11

Growth, Development & Milestones

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024,May 2025 (growth chart, stunting question confirmed)

Exam Frequency (Audited): Confirmed in virtually every session; milestone ages, red flags, regression, and growth assessment most tested

OVERVIEW

Growth and developmental milestones are foundational pediatrics. Every AIIMS/INI CET session includes at least one question on milestone ages, developmental red flags, screening tools, or growth chart interpretation. May 2025 confirmed a question on stunting with excessive weight gain (hypothyroidism scenario).

KEY FACTS FOR EXAM

- Gross motor: head control 3m, rolling 4-5m, sitting unsupported 6-7m, standing 9m, walking 12-15m, running 18m
- Fine motor: palmar grasp 3m, transferring objects 6m, pincer grasp 9-10m, scribble 15-18m, draw circle 3yrs
- Language: social smile 6 weeks, cooing 2m, babbling (consonants) 6m, mama/dada 9m, single words 12m, 2-word phrases 18m, sentences 3yrs
- Social: regards face 4 weeks, social smile 6 weeks, stranger anxiety 6-9m, parallel play 2yrs, cooperative play 4yrs
- Growth: weight doubles 5m, triples 1yr, quadruples 2yrs; height increases 25cm in year 1, 12cm in year 2
- Red flags: no social smile by 3m; no sitting by 9m; no single words by 18m; loss of acquired milestones (regression) = always pathological
- Denver Developmental Screening Test (DDST): screening tool; 4 domains; 90th percentile failure = delay
- M-CHAT: autism screening at 16-30 months; 6 items critical; 2+ critical = refer for diagnostic assessment

CLINICAL FEATURES

Stunting + excess weight	Height <-2SD + weight >+2SD + recurrent infections → suspect hypothyroidism (confirmed May 2025 question)
Regression	Loss of acquired milestones = always pathological; investigate for metabolic/degenerative disease
Isolated language delay	First investigate hearing (BERA/OAE); second = autism screen (M-CHAT)
Isolated motor delay	Investigate for CP, neuromuscular disease, hypothyroidism
Cerebral palsy	Non-progressive motor disorder; spastic diplegia most common in preterms; scissor gait
Autism spectrum	Triad: social communication deficit + restricted interests + repetitive behaviors

KEY INVESTIGATIONS

- Thyroid function: essential in all global developmental delay and growth failure
- Hearing assessment: BERA (brainstem evoked response audiometry) or OAE in language delay
- MRI brain: for regression, asymmetry, microcephaly, focal signs
- Metabolic workup: urine organic acids + plasma amino acids in regression

EXAM TIP

Confirmed May 2025: child with height <-2SD, weight >+2SD, recurrent respiratory infections (oral medications) → next best step = thyroid function tests (not 'promote physical activity'). Key milestones: social smile 6 weeks; pincer grasp 9-10 months; 2-word phrases 18 months. Regression = ALWAYS pathological. M-CHAT at 16-30 months for autism. Hand-wringing + regression in girls = Rett syndrome (MECP2).

Topic 12

Cystic Fibrosis

Verified Session Recall: INI CET Nov 2023 (sweat chloride question confirmed on slideshare recall); INI CET 2024 (confirmed collegedekho recall); AIIMS PG 2018,2019

Exam Frequency (Audited): Confirmed in recent sessions 2022-2025; CFTR mutation, sweat chloride paradox, and pancreatic involvement most tested

OVERVIEW

Cystic fibrosis is tested through its genetics, pathophysiology (CFTR mutation), and clinical consequences across organ systems. The classic exam question — sweat chloride vs pancreatic chloride — was directly confirmed in INI CET November 2023 recall papers. CF is increasingly tested as clinical vignettes in recent sessions.

KEY FACTS FOR EXAM

- Autosomal recessive; CFTR gene on chromosome 7q31; most common mutation = F508del (deltaF508 = phenylalanine deletion at position 508)
- CFTR: chloride channel on apical membrane; dysfunction leads to thick, viscid secretions in all exocrine glands
- Sweat gland: defective Cl reabsorption → HIGH sweat chloride (>60 mEq/L = diagnostic) — Na and Cl both high in sweat
- Pancreatic duct: defective Cl secretion → LOW pancreatic Cl → thick pancreatic secretions → exocrine pancreatic insufficiency
- Confirmed exam question (INI CET Nov 2023): sweat Cl increased + pancreatic secretion Cl decreased — answer = CF
- Lungs: chronic *Pseudomonas aeruginosa* colonization (most common and prognostically significant organism)
- Diagnosis: sweat chloride test (pilocarpine iontophoresis) + genetic testing; newborn screen (IRT + DNA)
- Treatment: CFTR modulators (ivacaftor, lumacaftor-ivacaftor, elexacaftor-tezacaftor-ivacaftor) for specific mutations

CLINICAL FEATURES

Respiratory	Chronic productive cough; recurrent pneumonia; bronchiectasis; Staph aureus early, <i>Pseudomonas</i> late
GI/Nutrition	Meconium ileus (neonatal); steatorrhea; failure to thrive; fat-soluble vitamin deficiency
Pancreas	Exocrine insufficiency (85%); recurrent pancreatitis; CF-related diabetes (CFRD) in adults
Liver	Biliary cirrhosis; fatty liver; portal hypertension in 5-10%
Reproductive	Males = obstructive azoospermia (absent vas deferens); fertility reduced in females
Nasal	Nasal polyps in children (think CF; rare in children without CF)

KEY INVESTIGATIONS

- Sweat chloride: >60 mEq/L = diagnostic; 30-60 = borderline; <30 = normal; test done twice for confirmation
- Genetic testing: confirm CFTR mutation; F508del most common (70% of CF alleles in Caucasians)
- Chest X-ray: hyperinflation; peribronchial thickening; bronchiectasis (upper lobe predilection)
- PFTs: obstructive pattern; FEV1 used to monitor progression and guide CFTR modulator eligibility

EXAM TIP

*Confirmed INI CET Nov 2023 question: sweat Cl INCREASED + pancreatic secretion Cl DECREASED = CF. This paradox is the #1 CF exam question: sweat = can't reabsorb Cl (high in sweat); pancreas = can't secrete Cl (low in pancreatic juice). Most common lung organism = *Pseudomonas aeruginosa* (late, chronic, biofilm). Most common overall pathogen early = *Staph aureus*. Meconium ileus = CF until proven otherwise.*

Topic 13

Dengue Fever in Children

Verified Session Recall: INI CET 2020,2021,2022,2023,2024,May 2025 (dengue + thrombocytopenia management question confirmed)

Exam Frequency (Audited): Confirmed in 80%+ of INI CET sessions; WHO 2009 classification, platelet transfusion criteria, fluid management most tested

OVERVIEW

Dengue is consistently tested in INI CET given its clinical burden in India. The 2009 WHO classification guides management. The May 2025 recall confirmed a question about dengue with platelet count 20,000 without bleeding — the answer was observation and fluid management (NOT platelet transfusion).

KEY FACTS FOR EXAM

- *Aedes aegypti* (day-biting); 4 serotypes (DENV 1-4); secondary infection with different serotype = more severe
- WHO 2009: Dengue without warning signs → Dengue with warning signs → Severe dengue
- Warning signs: abdominal pain, persistent vomiting, fluid accumulation, mucosal bleed, lethargy, liver >2 cm, rising HCT + rapid fall in platelets
- NS1 antigen: positive Days 1-5; IgM: from Day 5-6; IgG: rises rapidly in secondary infection
- Critical phase: Days 3-7 after fever onset; plasma leakage; shock (DSS); most dangerous period
- Platelet transfusion: ONLY if platelet <10,000/mm³ AND active significant bleeding; NOT for thrombocytopenia alone
- Confirmed May 2025: dengue Day 5, BP normal, platelets 20,000, NO bleeding → next step = monitor, fluid management (NOT platelet transfusion)
- Aspirin and NSAIDs: CONTRAINDICATED in dengue (risk of Reye's syndrome, bleeding, gastritis)

CLINICAL FEATURES

Febrile phase	High fever (39-40°C); severe headache; retro-orbital pain; myalgia; flushing; rash
Critical phase	Defervescence + plasma leakage → abdominal pain + pleural effusion + ascites + shock
Recovery phase	Fluid reabsorption; bradycardia; characteristic maculopapular rash; pruritis
Dengue rash	'Islands of white in sea of red'; maculopapular; appears in recovery phase
Tourniquet test	≥10 petechiae per 1 inch ² area = positive; positive in dengue (capillary fragility)
DSS (Dengue Shock Syndrome)	Cold extremities + narrow pulse pressure + restlessness; fluid bolus isotonic crystalloid

KEY INVESTIGATIONS

- NS1 antigen: Days 1-5 (primary infection); slightly shorter in secondary
- IgM ELISA: Day 5 onwards; IgG: secondary infection marker (rapid high rise)
- CBC: leukopenia; progressive thrombocytopenia; rising HCT (>20% rise = significant hemoconcentration)
- LFTs: AST often > ALT (dengue hepatitis); elevated in all cases

EXAM TIP

Confirmed May 2025: dengue + normal BP + platelet 20,000 + NO bleeding → answer = conservative management, NOT platelet transfusion. Platelet transfusion threshold = <10,000 + active bleeding. Fluid of choice in DSS = isotonic crystalloid (NS or RL); colloid if crystalloid fails. Aspirin = absolutely contraindicated. Tourniquet test ≥10 petechiae = positive. NS1 detectable Days 1-5; IgM from Day 5.

Topic 14

Thalassemia & Hemoglobin Disorders

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Confirmed in most sessions; HPLC pattern, chelation, PBS findings, and genetics most tested

OVERVIEW

Beta-thalassemia major is the prototype for exam purposes. Questions test genetics, blood film, HPLC pattern, transfusion targets, iron chelation, and complications. Alpha-thalassemia genetics (gene deletions) are also regularly tested. Sickle cell anemia features appear in the same topic block.

KEY FACTS FOR EXAM

- Beta-thal major: absent/reduced beta-globin; HbA absent; HbF very high; HbA2 elevated; needs transfusions from 6 months
- Pre-transfusion Hb target: 9-10.5 g/dL (suppresses ineffective erythropoiesis + extramedullary hematopoiesis)
- Iron overload: primary complication; heart (restrictive/dilated CMP) is leading cause of death; liver cirrhosis; DM; hypothyroidism
- Chelation: serum ferritin >1000 ng/mL → start chelation; deferoxamine (SC); deferasirox (oral, first-line); deferiprone (oral)
- HPLC: HbF high + HbA absent + HbA2 >3.5% (beta-thal major); HbA2 >3.5% + HbF slightly raised = beta-thal trait
- Alpha-thal: 2 deletions = trait (silent); 3 deletions = HbH disease (beta-4 tetramers); 4 deletions = Hb Barts (hydrops fetalis, incompatible with life)
- PBS in thal major: severe hypochromic microcytic anemia; target cells; nucleated RBCs; poikilocytosis; basophilic stippling
- Skull X-ray: 'hair-on-end' appearance (diploic space widening from marrow expansion); 'Erlenmeyer flask' femur

CLINICAL FEATURES

Pallor	Severe anemia; transfusion-dependent by 6 months of life
HSM	Massive hepatosplenomegaly (extramedullary hematopoiesis + iron overload); splenectomy if hypersplenism
Thalassemic facies	Maxillary hypertrophy; frontal bossing; depressed nasal bridge; 'chipmunk face'
Growth failure	Short stature; delayed puberty due to iron-induced hypogonadotropic hypogonadism
Cardiac	Dilated CMP + arrhythmias from iron overload; leading cause of death; T2* MRI monitors iron
Sickle cell	Vaso-occlusive crises; acute chest syndrome; dactylitis in infancy; splenic sequestration

KEY INVESTIGATIONS

EXAM TIP

- PBS: hypochromic microcytic; target cells; NRBCs; poikilocytosis; Howell-Jolly bodies (after splenectomy)
 - HPLC: gold standard; HbF, HbA, HbA2 quantification; detects thal, sickle cell, HbE, HbH
 - Serum ferritin + MRI liver T2*: iron overload monitoring (T2* <20 ms = significant liver iron)
 - Cardiac MRI T2*: <20 ms = significant cardiac iron → increase chelation intensity
- Key HPLC patterns: (1) HbA absent + HbF >90% + HbA2 >3.5% = beta-thal major. (2) HbA2 >3.5% + normal HbF = beta-thal trait (carrier). (3) HbH disease = 3 alpha gene deletions. (4) Hb Barts = 4 deletions = gamma-4 = hydrops fetalis. (5) PBS: target cells always present; NRBCs (nucleated RBCs) = hallmark of thal major. (6) Curative treatment = allogeneic HSCT; best outcomes Pesaro Class I (no organ damage).*

Congenital Hypothyroidism

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Confirmed in most sessions; NBS, posterior fontanelle sign, treatment window most tested

OVERVIEW

Congenital hypothyroidism is the most common preventable cause of intellectual disability (where NBS exists). Questions consistently test the NBS protocol, timing of treatment, target TSH, clinical signs (especially posterior fontanelle), and distinction from later-onset Hashimoto's thyroiditis.

KEY FACTS FOR EXAM

- Most common cause: thyroid dysgenesis (agenesis or ectopic thyroid) in 80-85% of cases
- NBS: TSH on heel-prick blood spot (Guthrie card) at Day 3-5; TSH >20 mIU/L = suspect CH
- Confirmatory: serum TSH (very high) + free T4 (very low); thyroid scan (Tc-99m pertechnetate)
- Most specific early sign: posterior fontanelle >5 mm at birth (normal PF closes by 6-8 weeks)
- Classic signs: prolonged jaundice + constipation + umbilical hernia + macroglossia + hoarse cry + hypotonia + dry skin
- Bone age: retarded; absent distal femur epiphysis at birth indicates severe/prolonged CH
- Treatment: L-thyroxine 10-15 mcg/kg/day; must start within 2 weeks of birth for normal neurological outcome
- Target TSH: 0.5-2 mIU/L in first 3 years (critical window for brain development); re-evaluate at 3 years

CLINICAL FEATURES

Neonatal period	Prolonged physiological jaundice (most common early sign); large PF; hypotonia
Facies	Coarse features; puffy periorbital; depressed nasal bridge; macroglossia; large tongue
Abdomen	Umbilical hernia; constipation; abdominal distension
Skin	Dry; cold; mottled skin; myxedema (non-pitting)
Neurological	Hypotonia; sluggish reflexes; lethargy; poor feeding; intellectual disability if untreated
Later childhood	Short stature; bone age retarded; delayed dentition; delayed puberty (Hashimoto's more common in this age group)

KEY INVESTIGATIONS

- NBS: TSH filter paper Day 3-5; if high → confirm with serum TSH + free T4
- Thyroid scan: detects ectopic thyroid, agenesis, goitre (ectopic most common in dysgenesis)
- Bone age X-ray: retarded; absent distal femoral epiphysis in severe CH at birth
- Anti-thyroid antibodies (anti-TPO): positive in Hashimoto's thyroiditis (NOT in congenital hypothyroidism)

EXAM TIP

Essential: (1) Most common preventable cause of intellectual disability = CH (if NBS available). (2) Posterior fontanelle >5 mm at birth = strongest clinical clue at birth. (3) Treatment must start within 2 weeks of birth. (4) Target TSH in first 3 years = 0.5-2 mIU/L (neurological window). (5) Hashimoto's = most common cause of acquired hypothyroidism in children >6 years (NOT congenital). (6) Pendred syndrome = sensorineural deafness + goitre + hypothyroidism (SLC26A4/pendrin mutation).

Topic 16

Acute Rheumatic Fever & Rheumatic Heart Disease

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Confirmed in most sessions; Jones criteria, Carey Coombs murmur, secondary prophylaxis most tested

OVERVIEW

ARF remains a major cause of acquired valvular disease in developing countries. Questions test the 2015 revised Jones criteria, Sydenham's chorea recognition, the Carey Coombs murmur, and secondary prophylaxis duration. The distinction between ARF criteria for high-risk vs low-risk populations is also tested.

KEY FACTS FOR EXAM

- Causative: Group A beta-hemolytic Streptococcus (GAS) pharyngitis ONLY (NOT skin infection); molecular mimicry pathogenesis
- Evidence of GAS: positive throat culture OR positive RADT OR elevated/rising ASO titre
- 2015 Jones Criteria: 2 major OR 1 major + 2 minor + evidence of preceding GAS infection
- Carditis: pancarditis; mitral valve most commonly affected (mitral regurgitation first, then MS later)
- Major criteria: Carditis (including subclinical on echo), Polyarthritides/Arthralgia (high risk), Chorea, Erythema marginatum, Subcutaneous nodules
- Carey Coombs murmur: soft mid-diastolic rumble due to acute mitral valve inflammation (mitral valvulitis)
- Minor criteria: Fever $>38.5^{\circ}\text{C}$, elevated ESR/CRP, prolonged PR interval
- Secondary prophylaxis: benzathine penicillin G 1.2 MU IM every 3-4 weeks; duration based on carditis severity

CLINICAL FEATURES

Carditis	Most prognostically important; new murmur; tachycardia disproportionate to fever; MR murmur
Polyarthritides	Most common presentation; migratory; large joints (knees, ankles, elbows, wrists); exquisitely tender; responds dramatically to aspirin
Chorea	Purposeless involuntary movements; 'milkmaid grip'; emotional lability; 3-6 months after pharyngitis; isolated chorea = sufficient for ARF diagnosis
Erythema marginatum	Transient; non-pruritic; central clearing; pink/red margins; trunk and proximal limbs; NOT face
Subcutaneous nodules	Firm; painless; over bony prominences; associated with severe carditis only
ASO titre	Rises 3-4 weeks after GAS infection; >200 Todd units = significant; anti-DNAse B more sensitive

KEY INVESTIGATIONS

- ASO titre: rises 3-4 weeks post-pharyngitis; most commonly used test; >200 Todd units significant
- Anti-DNAse B: more specific and remains elevated longer than ASO; positive in skin infections too
- Echo: subclinical carditis (Doppler shows valve regurgitation) — counts as major criterion in 2015 Jones
- CRP/ESR: elevated; normalize as inflammation resolves; used for monitoring

EXAM TIP

Critical points: (1) Chorea alone = sufficient for ARF diagnosis (does not need other criteria or evidence of GAS). (2) Carey Coombs murmur = soft mid-diastolic due to mitral valve inflammation. (3) 2015 update: subclinical carditis on echo counts as major criterion. (4) Arthritis in ARF responds dramatically to aspirin (if good response = supports diagnosis). (5) Secondary prophylaxis duration: no carditis = 5 years; carditis no residual = 10 years; carditis with residual = lifelong.

Henoch-Schonlein Purpura (IgA Vasculitis)

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Confirmed in most sessions; normal platelets, intussusception complication, IgA deposits most tested

OVERVIEW

HSP (now officially IgA Vasculitis) is the most common systemic vasculitis in children. The key exam hook is that despite widespread purpura the platelet count is **NORMAL**, distinguishing it from ITP. The renal biopsy showing IgA deposits, intussusception complication, and steroid indications are classic questions.

KEY FACTS FOR EXAM

- IgA immune complex deposition in small vessel walls; same IgA pathology as IgA nephropathy (Berger's disease)
- Classic tetrad: palpable purpura + arthritis/arthralgia + abdominal pain + renal involvement
- PLATELET COUNT NORMAL** — most important differentiating feature from ITP (thrombocytopenic purpura)
- Rash: non-thrombocytopenic palpable purpura; gravity-dependent; lower limbs + buttocks; non-blanching
- Renal involvement: 30-50%; hematuria ± proteinuria; HSP nephritis can progress to CKD in severe cases
- Intussusception: most common surgical complication; ileo-ileal type (NOT ileo-colic as in idiopathic); US → reduction
- Steroids: reduce GI pain and joint symptoms but do NOT prevent nephritis; NOT for rash alone
- Diagnosis: EULAR/PRINTO criteria: purpura/petechiae + ≥1: arthritis, GI pain, renal involvement, IgA deposition on biopsy

CLINICAL FEATURES

Purpura	Palpable (raised), non-blanching; starts on buttocks/legs; crops over days-weeks; may extend to arms
Arthritis	Periarticular swelling; large joints (knees, ankles); painful but no permanent damage
Abdominal pain	Colicky; precedes rash in 30%; guaiac-positive stools; intussusception if severe
Renal	Hematuria (most common); proteinuria; nephrotic-nephritic overlap possible; check urine for 6 months
Scrotal	Scrotal edema/orchitis in boys; must differentiate from testicular torsion (Doppler USS)
Angioedema	Periorbital, scalp, dorsum of hands — non-pitting; distinct from cardiac edema

KEY INVESTIGATIONS

- Urinalysis: hematuria + casts + proteinuria — most important monitoring investigation
- Platelets: **NORMAL** — if low, reconsider diagnosis (ITP, meningococcemia, TTP)
- Serum IgA: elevated in ~50% (not diagnostic; can be normal)
- Renal biopsy: IgA mesangial deposits; indicated for nephrotic-range proteinuria or rapidly progressive renal disease

EXAM TIP

*Most tested points: (1) **PLATELETS NORMAL** = differentiates from ITP. (2) Most common surgical complication = intussusception (ileo-ileal, not ileo-colic). (3) Steroids: treat GI and joint pain; do NOT prevent nephritis. (4) Renal biopsy: IgA mesangial deposits = same as Berger's IgA nephropathy. (5) Monitor urine weekly for 6 months post-resolution — nephritis can appear late. (6) Blood pressure monitoring mandatory.*

Neurocutaneous Syndromes (Phakomatoses)

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024

Exam Frequency (Audited): Confirmed in most sessions; TSC ash-leaf spot, NF1 diagnostic criteria, Sturge-Weber calcification most tested

OVERVIEW

Neurocutaneous syndromes are classic AIIMS-style topics due to their high factual density. Tuberous sclerosis, neurofibromatosis types 1 and 2, and Sturge-Weber syndrome are the four most commonly tested. Questions focus on pathognomonic features, gene mutations, and inheritance patterns.

KEY FACTS FOR EXAM

- Tuberous Sclerosis (TSC): TSC1 (hamartin, chr 9) or TSC2 (tuberin, chr 16); autosomal dominant; mTOR pathway
- TSC skin: ash-leaf macule (earliest — present at birth, UV lamp); facial angiofibromas (erroneously called 'adenoma sebaceum'); Shagreen patch; periungual fibromas (Koenen's tumors)
- TSC brain: cortical tubers → seizures; subependymal nodules → subependymal giant cell astrocytoma (SEGA); infantile spasms
- NF1: NF1 gene (chr 17q11.2); diagnostic criteria = ≥ 2 of 7: ≥ 6 cafe-au-lait spots, ≥ 2 neurofibromas, ≥ 1 plexiform, axillary/inguinal freckling, optic glioma, ≥ 2 Lisch nodules, 1st degree relative
- NF2: NF2 gene (chr 22q12); bilateral vestibular schwannomas (acoustic neuromas); meningiomas; ependymomas
- Sturge-Weber: port-wine stain (naevus flammeus) in V1 trigeminal distribution; ipsilateral leptomeningeal angioma; glaucoma; 'tram-track' calcification on CT
- Ataxia-Telangiectasia: AR; ATM gene; cerebellar ataxia + conjunctival telangiectasia + IgA/IgE deficiency + malignancy risk
- mTOR inhibitors (everolimus): treatment for SEGA in TSC; also for angiomyolipoma

CLINICAL FEATURES

TSC seizures	Infantile spasms (West syndrome) most common in infancy; later Lennox-Gastaut pattern
Ash-leaf macule	Hypopigmented oval macule; earliest TSC sign; visible at birth; enhanced by Wood's lamp (UV)
Lisch nodules	Melanocytic iris hamartomas; pathognomonic for NF1; slit-lamp examination required
Port-wine stain	Capillary vascular malformation; V1 distribution; does NOT fade with age; laser treatment
AT features	Cerebellar ataxia + conjunctival telangiectasia + low IgA/IgE + elevated AFP + lymphoma/leukemia risk
Plexiform neurofibroma	Pathognomonic for NF1; risk of malignant transformation to MPNST (malignant peripheral nerve sheath tumor)

KEY INVESTIGATIONS

EXAM TIP

- MRI brain: cortical tubers (TSC); leptomeningeal enhancement (Sturge-Weber); acoustic neuroma (NF2); SEGA
 - CT head: 'tram-track' calcifications (Sturge-Weber); subependymal calcified nodules (TSC)
 - Genetic testing: TSC1/TSC2, NF1, NF2, ATM mutations
 - Ophthalmology: Lisch nodules (slit-lamp); glaucoma (Sturge-Weber); retinal hamartomas (TSC)
- Key mnemonics: NF1 criteria = 6-2-1-2-1 (6 cafe-au-lait, 2 neurofibromas, 1 plexiform, 2 Lisch, 1 relative). NF2 = bilateral = chromosome 22. TSC = 2 chromosomes (9 and 16). Ash-leaf macule = earliest TSC sign. 'Tram-track' = Sturge-Weber. AT: 'ataxia + glasses (telangiectasia)'; very elevated AFP; IgA/IgE deficient. mTOR inhibitors for SEGA in TSC. Sturge-Weber = NOT inherited (sporadic; GNAQ mutation).*

Topic 19

Vaccination Schedule & National Immunization Programme

Verified Session Recall: AIIMS PG 2015,2016,2017,2018,2019; INI CET 2020,2021,2022,2023,2024,2025 — confirmed every session

Exam Frequency (Audited): Confirmed in EVERY session; schedule timing, contraindications, cold chain, AEFI most tested

OVERVIEW

Immunization is tested in every AIIMS PG and INI CET paper. The IAP ACVIP schedule is updated every 1-2 years and differences from the UIP schedule are favourite exam questions. Cold chain management, vaccine contraindications, and AEFI categorization are also consistently tested.

KEY FACTS FOR EXAM

- At birth: BCG + OPV-0 + Hepatitis B (within 24 hours of birth, ideally within 12 hours for HBV)
- 6 weeks, 10 weeks, 14 weeks: DTwP/DTaP + IPV + Hib + Hepatitis B + PCV + Rotavirus
- 9 months: MR (Measles-Rubella); 12 months: Hepatitis A (dose 1); 15 months: MMR + Varicella + PCV booster
- Live vaccines: BCG, OPV, MMR, Varicella, Rotavirus, Yellow fever, Oral typhoid — AVOID in immunocompromised
- OPV: NOT in HIV-positive children (risk of VAPP); use IPV instead
- Cold chain: most vaccines 2-8°C; OPV at -20°C (freeze); BCG and MMR are light-sensitive
- VVM (vaccine vial monitor): heat-sensitive; if inner square = darker than outer ring → discard
- Shake test: freeze-sensitive vaccines (DPT, Hep B, TT); if clumping → discarded (freeze-damaged)

CLINICAL FEATURES

BCG reaction	Papule at 2 weeks → ulcer → scar by 3 months; BCGitis = lymphadenitis in immunodeficient
VAPP	Vaccine-associated paralytic polio: from OPV (Sabin strains); type 3 most common; replaced by IPV in India UIP now
Anaphylaxis	Within 15-30 min of vaccine; adrenaline 0.01 mg/kg IM (thigh) immediately
AEFI categories	Vaccine reaction / Program error / Coincidental / Injection reaction / Unknown
Seroprotection	Anti-HBs >10 IU/L = protected after hepatitis B vaccination
AFP surveillance	Any <15 years with acute flaccid paralysis → report within 24 hours to surveillance system

KEY INVESTIGATIONS

EXAM TIP

- Post-vaccination serology: anti-HBs >10 IU/L for HBV; check in high-risk groups
 - AEFI reporting: all serious AEFIs to national surveillance within 24-48 hours
 - Cold chain verification: temperature monitoring; log tag; VVM on every vial
- Most tested: (1) Birth vaccines = BCG + OPV-0 + Hep B. (2) Live vaccines = BCG, OPV, MMR, Varicella, Rotavirus, YF, Oral Typhoid — contraindicated in immunocompromised and pregnancy. (3) OPV contraindicated in HIV+ children. (4) VVM darker inside = discard. (5) HPV vaccine: girls 9-14 years; 2 doses if <15 years; 3 doses if ≥15 years. (6) IAP 2023: annual influenza vaccine from 6 months; meningococcal vaccine for high-risk; Hep A at 12 and 18 months.*

Topic 20

Neonatal Respiratory Distress Syndrome (RDS) & Neonatal Infections

Verified Session Recall: INI CET May 2025 (neonatal RDS confirmed); INI CET 2021,2022,2023,2024; AIIMS PG 2015-2019

Exam Frequency (Audited): Confirmed in 80%+ of sessions; RDS vs TTN vs MAS differentiation, surfactant therapy, neonatal sepsis most tested

OVERVIEW

Neonatal respiratory distress is a core neonatal topic covering RDS (preterm, surfactant deficiency), Transient Tachypnea of the Newborn (TTN), Meconium Aspiration Syndrome (MAS), and neonatal sepsis. The May 2025 INI CET confirmed an RDS question. These conditions require clinical differentiation and specific management.

KEY FACTS FOR EXAM

- RDS (Hyaline Membrane Disease): preterm <34 weeks; surfactant deficiency; type II pneumocyte immaturity; onset at birth
- CXR in RDS: ground-glass (reticulogranular) opacity + air bronchograms; lung volumes reduced (bell-shaped)
- Surfactant: poractant alfa (Curosurf) or beractant (Surfanta); give endotracheally; repeat doses if needed
- Antenatal corticosteroids (betamethasone): given to mother 24-34 weeks; accelerates fetal lung maturity; reduces RDS severity
- TTN: term/near-term; delayed clearance of fetal lung fluid; onset at birth; CXR shows wet/streaky perihilar opacities; resolves in 24-48h
- MAS: term/post-term; meconium inhalation; 'snowstorm' CXR; air-leak (pneumothorax) risk; risk of PPHN
- Management of MAS: do NOT suction routinely (2015 guidelines removed routine suctioning); intubate only if not vigorous
- Neonatal sepsis: early onset (<72h) = GBS, E. coli, Listeria; late onset (>72h) = Staph epidermidis, Klebsiella, Enterococcus

CLINICAL FEATURES

RDS signs	Grunting; nasal flaring; tachypnea (>60/min); subcostal/intercostal retractions; cyanosis; worsens first 48-72h
TTN signs	Tachypnea onset at birth; resolves by 24-48h; mild; O2 requirement; risk factors: cesarean section, diabetic mother
MAS signs	Meconium-stained liquor; barrel-shaped chest (air trapping); hypoxia disproportionate to CXR
Grunting	Physiological PEEP by partial glottis closure; maintains FRC; universal sign of neonatal lung disease
CLD (BPD)	Bronchopulmonary dysplasia: oxygen-dependent at 36 weeks PCA; complication of prematurity + mechanical ventilation
Neonatal sepsis	Temperature instability + poor feeding + lethargy + apnea + abdominal distension

KEY INVESTIGATIONS

- CXR: RDS = reticulogranular + air bronchograms; TTN = wet perihilar; MAS = patchy irregular; pleural effusion
- ABG: hypoxemia + hypercapnia (RDS, MAS); normal CO2 (TTN); metabolic acidosis (sepsis)
- Gastric aspirate shake test: foam = surfactant present (mature lungs); no foam = immature lungs (will develop RDS)
- Blood culture: essential in neonatal sepsis before antibiotics; CRP; PCT; CSF if stable

EXAM TIP

Confirmed May 2025 question on neonatal RDS (preterm with severe respiratory distress). Key differentials: (1) RDS = preterm, onset at birth, worsens 48-72h, air bronchograms on CXR. (2) TTN = term, onset birth, resolves 24-48h, wet perihilar opacities. (3) MAS = term/post-term, meconium liquor, air trapping on CXR. (4) Antenatal betamethasone = reduces RDS (given 24-34 weeks gestation). (5) Suctioning in MAS: intubate + suction ONLY if not vigorous (not routine for all meconium-stained babies since 2015).